

Return-Weighted Risk for Navigating an Evolving AI Landscape

Reconnecting Risk with Value in AI Risk Management

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Abstract

Organizations increasingly govern AI activities while still learning what outcomes those activities produce, what value those outcomes create, what resources are required to realize that value, and what risks the activities introduce. Rapid changes in capabilities, costs, uses, competitors, controls, and operating conditions can make even careful point-in-time assessments short-lived. We propose Return-Weighted Risk (RWR), an activity-level governance approach that keeps the value case connected to an independently assessed risk case as both evolve. Authorization depends on three sequential tests: admissibility, risk permissibility, and value justification. Material change can reopen the decision. Independent risk identification provides the coverage floor; intended value adds purpose-directed emphasis to additional risk inquiry. RWR therefore shifts the governance question from whether value and risk are each being managed to whether current evidence about both still supports the same organizational action. A hypothetical small-business-lending case illustrates the logic. We position RWR alongside established enterprise, banking, benefits, adaptive, and AI-governance approaches and identify empirical questions for later validation.

Keywords: AI governance; AI risk management; return-weighted risk; activity-level governance; concurrent learning; continuing authorization; material change; value realization

1. The Governance Problem in an Evolving AI Environment

Organizations adopt artificial intelligence because they expect beneficial change—financial, operational, customer-related, workforce-related, strategic, scientific, public, or informational. The same activities can create or amplify risks involving reliability, privacy, security, fairness, compliance, model behavior, third-party dependence, operational resilience, and human reliance. What makes the present environment especially difficult is that organizations often have to learn about value and risk while the activity is already being developed, piloted, deployed, and changed.

That creates a moving governance target. AI capabilities, deployment configurations, user behavior, available alternatives, and organizational objectives can change before the next decision point. A sound assessment today may therefore lose decision relevance sooner than governance processes built around more stable assumptions would normally expect.

When both sides are unsettled, the familiar intuition that risk can first be characterized and then used to condition a return decision becomes harder to apply. For some AI initiatives, the organization does not yet

know enough about either the durable value pathway or the realized risk profile to treat one as a stable weight on the other. Waiting for complete knowledge can prevent useful learning; proceeding as though the initial business and risk cases were settled can allow authorization to persist after its justification has changed.

1.1 A motivating decision: Aster Vale

Aster Vale Regional Bank has completed a six-month pilot of Navigator, a vendor-developed generative-AI system used to organize documents and draft portions of credit memoranda for small-business loans. Navigator does not approve or decline applications, assign credit ratings, determine prices, or communicate directly with applicants. Underwriters retain formal decision authority and certify every completed memorandum.

The pilot appears successful from one perspective. Navigator reduces underwriter preparation time by 28 percent, experienced underwriters report spending less time sorting documents, and the system remains within its restricted scope. The bank identifies no autonomous credit decision, serious privacy incident, or explicit breach of its approved pilot risk limits.

The broader evidence is mixed. Application-to-decision time declines only 7 percent because document preparation is not the principal source of many delays. Repeated requests for applicant information fall only slightly, abandonment scarcely changes, and the anticipated improvement in credit access is not clearly established. A non-AI workflow redesign produces a similar improvement in end-to-end processing time.

Navigator performs best with standardized digital information; scanned, handwritten, multilingual, or unconventional records require more manual work. Some newer underwriters increasingly begin with Navigator summaries, and source accessibility deteriorates. The bank gains useful knowledge about generative AI while becoming more dependent on a vendor whose system it understands only partially.

Aster Vale management nevertheless proposes expansion to all small-business lending teams and larger loans. Its new rationale emphasizes processing capacity, avoided hiring, and strategic AI capability. The pilot, however, had originally been justified mainly by faster decisions, lower applicant burden, and improved access. The revised rationale may be legitimate, but it is materially different.

The expansion decision facing Aster Vale is therefore straightforward to state: given what the bank has now learned about Navigator's value and risk, does the evidence still support broader deployment?

1.2 Research question and proposition

We ask how organizations should govern AI activities when they must learn about value realization and risk realization at the same time that they are deciding whether to continue, adapt, expand, restrict, suspend, or terminate those activities.

Our proposition is that this setting calls for an explicit connection between two evolving bodies of evidence: the value case and an independently assessed risk case. RWR makes that relationship a condition of continuing authorization.

1.3 Contribution and organization

The contribution is a shift in the primary governance question. Instead of asking only whether value and risk are each being managed, RWR asks whether the current evidence about both still supports the same organizational action. Four features make that shift concrete: intended value helps direct additional risk inquiry without determining the risk conclusion; risk permissibility alone cannot justify continuation; feasible alternatives remain relevant after initial approval; and material change can reopen the authorization decision.

Section 2 explains why rapid AI evolution makes point-in-time assessments unusually transient. Section 3 positions the problem against existing governance foundations and identifies what remains insufficiently explicit. Sections 4 through 6 introduce RWR in plain language and develop its authorization and learning logic. Section 7 revisits Aster Vale as a governance decision. Section 8 discusses contribution, institutional implications, limitations, and research questions. Section 9 concludes.

2. Why AI Makes the Problem More Acute

Several features of today's AI environment make point-in-time assessment unusually fragile. They concern the speed of technological change, evolving purpose and system behavior, difficulty attributing value and risk, and the limited portability of evidence across settings and scale.

2.1 Rapid change makes point-in-time assessment transient

AI changes not only what organizations know, but how long that knowledge remains decision-relevant. A point-in-time (PIT) assessment may be carefully performed and still age quickly as model capabilities expand, vendors update systems, deployment patterns change, new uses emerge, costs shift, and competing AI or non-AI alternatives improve. The consequence applies to both sides of the decision. Expected and realized value can change because new capabilities or alternatives alter what is feasible and worthwhile; risk can change because new capabilities, configurations, users, and operating contexts create different exposures. RWR is motivated partly by this temporal problem: continuing governance cannot assume that a sound initial return-risk assessment remains stable simply because it was sound when made.

In April 2026, U.S. banking agencies explicitly excluded generative and agentic AI from revised model-risk-management guidance because those technologies were considered “novel and rapidly evolving” [2]. The agencies also announced plans for further information gathering focused on banks’ use of AI, including generative and agentic systems.

This does not show that established governance principles are unsuitable for AI; banks are still expected to apply appropriate internal governance and controls. It does illustrate the difficulty of assuming that a conventional point-in-time model-risk framework can fully settle governance for a rapidly changing AI activity.

2.2 Purpose and use can evolve

An AI system may move from document preparation to recommendation, from a restricted user group to an enterprise workflow, or from an internal aid to customer-facing interaction. Formal scope changes may be visible, but effective influence can change earlier through user reliance, workflow design, prompts, retrieval, tools, or vendor updates. Purpose therefore can drift before governance labels change.

2.3 System behavior is distributed

The behavior of a deployed AI activity depends on more than a model. Data, retrieval, configuration, prompts, tools, human review, vendor changes, and operating context jointly influence outcomes. Evidence from a pilot therefore concerns a configured activity under particular conditions, not an abstract model whose properties necessarily travel to a new setting.

2.4 Value attribution is difficult

An observed outcome is not automatically value. Task-level improvement may fail to produce workflow or stakeholder value if the improved task is not the binding constraint, if complementary work absorbs the gain, or if the outcome does not advance the relevant organizational objective. Faster document preparation, for example, may reveal that another process is the principal delay. Observed improvement may arise from training, process redesign, user selection, policy changes, AI use, or interactions among them. OECD evidence that many firms struggle to estimate AI return on investment is consistent with this difficulty [8]. Observed outcomes should therefore not be treated as realized value until their relevance to the stated objective is supported, and claims about AI contribution should remain proportionate to the available attribution evidence.

2.5 Risk attribution and distribution are also difficult

Risk may arise from model behavior, data, retrieval, workflow, human reliance, third parties, or interactions among them. Average performance can also conceal differences across users, document types, populations, and scales. The same system may reduce work for experienced employees while increasing verification burden for others, or improve standardized cases while worsening nonstandard ones. Distribution therefore belongs in both the value and risk cases.

2.6 Evidence may not travel

Scaling changes users, volumes, incentives, oversight, data, consequences, and sometimes the system itself. Evidence sufficient to continue a restricted pilot may be insufficient to expand its function or scale. This portability problem affects value evidence and risk evidence simultaneously.

2.7 Governing while learning

The resulting challenge is not that AI risk cannot be measured. It is that organizations may have to decide how to proceed while both value realization and risk realization remain only partially understood.

Governance becomes less a problem of optimizing known quantities and more a problem of learning about the relationships among action, value, and risk while exposure continues.

The situation resembles crossing a river by testing the footing ahead. The destination may be understood even when the complete path is not, and each controlled step produces information for the next. The analogy does not justify blind experimentation or unlimited exposure: progress must remain limited, reversible where possible, and directed toward a stated purpose. RWR adopts this logic of disciplined progression under incomplete and changing knowledge.

The governance implication is straightforward: controlled progression must be paired with explicit opportunities to re-decide.

Organizations do not face this problem without tools. Existing governance and decision frameworks already provide much of the foundation for managing value, risk, and learning; the question is how those pieces connect at the activity-level authorization decision.

3. Existing Foundations and the Remaining Governance Problem

RWR is not built from wholly new ingredients. Existing research and practice already offer important pieces of the approach across enterprise risk management, banking supervision, AI governance, benefits realization, adaptive management, benefit-risk decision making, and responsible innovation. Together, these fields show how organizations can govern value, govern risk, and learn as conditions change. Our question is how those capabilities should come together when a specific AI activity must be re-decided while both its value and risk evidence are evolving.

3.1 Enterprise and AI governance

Enterprise governance provides an important starting point. The Committee of Sponsoring Organizations of the Treadway Commission (COSO) enterprise risk management framework integrates risk with strategy and performance [21]. Banking provides a particularly developed sector-specific example. Office of the Comptroller of the Currency (OCC) new-activity guidance connects business strategy, risks and returns, performance criteria, alternatives, exit strategies, and ongoing review [1]. Revised model-risk and third-party guidance adds expectations for monitoring, validation, accountability, and continuing fitness for purpose [2], [3].

The National Institute of Standards and Technology Artificial Intelligence Risk Management Framework (NIST AI RMF) likewise asks organizations to establish context, intended purpose, beneficial uses, assumptions, limitations, and positive and negative impacts while managing risk across the lifecycle [5].

Organizational AI-governance scholarship also explicitly treats governance as necessary both to realize AI benefits and manage AI risks, and defines governance as a system of rules, practices, processes, and technological tools aligning AI use with organizational strategies, objectives, values, legal requirements, and ethical principles [22].

A subsequent systematic review characterizes organizational AI governance as a rapidly growing but fragmented field and identifies limited understanding of implementation and insufficient operationalization of governance processes as important knowledge gaps [23]. Design-oriented work has likewise proposed meta-requirements for practice-oriented organizational AI-governance frameworks [24]. Financial-sector surveys report both expected benefits and substantial concerns around data, privacy, security, bias, model complexity, third-party dependence, and skills [6], [7]. OECD evidence is especially relevant to the present motivation: firms report difficulty estimating AI return on investment, and AI projects often involve experimentation with no guarantee of success [8].

3.2 Benefits realization management and adaptive management

Benefits realization management (BRM) provides an established approach for maintaining accountability for expected value over time. Official guidance calls for benefits to be specified, assigned to owners, measured against baselines, monitored beyond project delivery, and reconciled with the business case [9], [10]. Empirical work nevertheless documents difficulties involving ownership, resources, and accountability [11]. Adaptive management provides an equally important precedent for acting while learning: stakeholders, objectives, alternatives, models, monitoring, and observed responses are used to update decisions [12]. Later work extends uncertainty to objective functions themselves and distinguishes technical learning from reconsideration of institutional decision elements [13], [14]. Dynamic Adaptive Policy Pathways and real-options reasoning add staged commitment, adaptation triggers, and options to expand or abandon [15], [16].

3.3 Benefit-risk constraints and responsible innovation

Benefit-risk regulation demonstrates that decisions under uncertainty need not collapse benefits and harms into an unconstrained trade. Medical-device regulation can consider uncertainty while maintaining safety and effectiveness requirements [17]. Tolerability-of-risk reasoning recognizes exposures that are not ordinarily justified and continuing duties to reduce risk [18]. Multicriteria benefit-risk research warns that simple aggregation can produce counterintuitive recommendations [19]. Responsible-innovation scholarship adds anticipation, reflexivity, inclusion, and responsiveness, extending attention beyond organizational return alone [20].

3.4 Relationship to existing approaches

RWR builds on several established and recent developments rather than beginning from a blank slate. Its components are familiar in different combinations across enterprise and AI governance, benefits realization management, adaptive management, benefit-risk decision making, responsible innovation, and recent work on dynamic value-risk tensions.

First, RWR treats value and risk as parts of the same governance conversation. This is consistent with enterprise risk management, banking new-activity governance, organizational AI governance, and the Paradox-based Responsible AI Governance (PRAIG) framework, all of which connect organizational objectives or value creation with risk and governance [1], [21]-[25]. RWR therefore does not claim that bringing value into risk governance is itself new.

Second, RWR assumes that governance must continue as evidence changes. Benefits realization management maintains accountability for expected benefits after approval, while adaptive management updates action as evidence emerges [9]-[16]. RWR extends this logic by asking whether changes in either the value case or the independently assessed risk case should change the authorization of a specific activity.

Third, RWR allows controlled action to generate decision-relevant learning. This is aligned with adaptive management, real-options reasoning, and Dynamic Adaptive Policy Pathways [12], [15], [16]. RWR applies that staged-learning logic to AI activity authorization: the exposure required to learn must itself remain admissible, risk-permissible, and justified by the expected decision value of learning.

Fourth, RWR does not treat value as compensation for every form of risk. Benefit-risk regulation and tolerability-of-risk approaches already distinguish between tradeable considerations and constraints that should not be overridden by expected benefit [17]-[19]. RWR uses a related separation by distinguishing admissibility and risk permissibility from value justification.

PRAIG is especially close to RWR in treating value creation and risk mitigation as a dynamic relationship rather than a static trade-off [25]. The difference is principally one of governance focus. PRAIG addresses organizational or portfolio posture toward persistent value-risk tensions; RWR focuses on the continuing authorization of a named activity as its value and risk evidence evolve. The two approaches can therefore be complementary.

Taken together, these workstreams supply much of the intellectual foundation for RWR. The distinctive move is to reorganize those elements around one activity-level question: whether the current value case and current risk case, considered relative to feasible alternatives, still support the same organizational action.

3.5 The remaining governance problem

Consider an AI activity that has already been authorized. Its residual risk may remain within approved limits and it may be producing some favorable outcomes, yet the value of those outcomes may differ from what was originally expected. A feasible alternative may have improved, the activity may have changed, or pilot evidence may no longer support the proposed scale. The organization still has to decide whether to continue, adapt, expand, restrict, suspend, or terminate the activity.

Existing approaches illuminate different parts of that decision. Risk management can assess whether exposure remains permissible. BRM can ask whether expected benefits are materializing. Adaptive management can identify what should be learned next. AI governance can address alignment, accountability, controls, and lifecycle obligations. PRAIG can frame the continuing tension between value creation and risk mitigation. What remains less explicit is the decision that joins these perspectives: do the current value case and current risk case, relative to feasible alternatives, still support the same authorized action?

RWR is proposed as an approach to this remaining problem. Section 4 explains how it keeps the evolving value case and independently assessed risk case connected to the continuing authorization of a specific AI activity.

4. Return-Weighted Risk: The Governance Approach

4.1 RWR in plain language

In practice, RWR starts from a simple situation. An organization approved an AI activity for a reason. As the activity operates, the organization learns what it actually delivers and what risks it actually creates. RWR asks whether the current value story and the current risk picture still support the same action. If either changes materially, the organization re-decides rather than merely continuing to monitor.

4.2 Why “Return-Weighted Risk”? The deliberate reversal

The contrast is simple: risk-weighted return conditions the evaluation of return on risk; RWR uses intended value to direct additional risk inquiry while keeping authorization tied to both value and independently assessed risk. The name deliberately reverses the familiar direction of risk-adjusted or risk-weighted return. In conventional investment reasoning, once risk and expected return are characterized sufficiently for comparison, the question is how much return is being obtained for the risk undertaken. RWR addresses an earlier and less settled governance condition, in which the value-producing pathway, its required inputs and costs, and the risk profile may all still be evolving. The reversal is conceptual, not mathematical.

Risk and return are not collapsed into a common score, and expected value cannot alter an independently assessed risk conclusion. Within RWR, the operational object on the economic side of the decision is the value case: the structured argument about the outcomes sought, their value, the resources and costs required to realize that value, and feasible alternatives. The word return in RWR retains the deliberate link to risk-weighted return and signals this economic side of the decision; it should not be read as a claim that all relevant value can already be reduced to a single financial return measure. Independent risk identification establishes the organization's coverage floor. The value case then adds purpose-directed emphasis toward dependencies and consequences that matter to the intended value pathway.

4.3 The value case

The value case is the structured argument for undertaking the activity; value evidence consists of the data, observations, and analysis that support or challenge that argument. It distinguishes outcomes from value: an observed change becomes relevant to the value case when its relationship to the stated objective, affected stakeholders, and feasible alternatives is sufficiently supported for the decision at hand.

For each proposed or authorized action, the value case records the current purpose; affected stakeholders; baseline and feasible alternatives; intended and observed outcomes; the value assigned to relevant outcomes; complementary inputs and expected costs required to realize that value; decision horizon; and supporting evidence. These are qualitative or quantitative, evidence-graded fields rather than necessarily commensurable numerical variables. Across the lifecycle, the value case distinguishes intended value, expected value, observed outcomes, realized value supported by evidence, and updated expected value. Attribution is a property of the evidence and reasoning supporting claims about what contributed to realized value, not a separate lifecycle state.

4.4 The risk case

The risk case is the structured assessment of the activity's risk; risk evidence consists of the information used to identify, estimate, control, and challenge that assessment. RWR uses risk in the narrower sense of possible adverse effects under uncertainty. Known or expected resource costs of undertaking the activity belong in the value/return analysis rather than being relabeled as risk; uncertainty about those costs can itself create risk.

The risk case records identified exposures and affected parties; likelihood and severity where assessable; uncertainty; controls; residual risk; permissible limits; and supporting evidence. Candidate legal, ethical, policy, or other non-compensable issues are flagged to the admissibility test rather than adjudicated as ordinary residual risk. Where feasible alternatives remain live, the risk analysis also includes enough comparative information to judge whether an alternative would materially change exposure; a full parallel risk case is not required for every alternative. Independent baseline risk identification is supplemented—but never replaced—by value-directed inquiry.

4.5 Coupled governance

Business owns the proposed value case and remains accountable for realization, but ownership does not confer unilateral authority to validate attribution or revise the rationale without challenge. Risk functions assess and challenge risk independently; they do not determine organizational purpose. An existing authorization body—or a governance process layered onto an existing committee—joins the two cases and decides whether both support the proposed action. RWR does not require creation of a new committee where suitable authority already exists.

5. Continuing Authorization Under RWR

RWR becomes operational through a continuing authorization decision. The organization does not treat approval as a one-time gate followed by passive monitoring; it asks whether the activity should remain authorized under current knowledge and conditions.

5.1 Three tests

RWR uses three sequential tests for the authorization decision. Their formal questions appear below, and Table 1 summarizes the plain-language decision logic and consequence of failing each test.

- **Admissibility:** Does the action satisfy applicable legal, ethical, policy, and non-compensable constraints?
- **Risk permissibility:** Does independently assessed residual risk remain within authorized limits for the proposed scope and conditions?
- **Value justification:** Is the current value case—including relevant outcomes, realized or expected value, required resources and costs, and feasible alternatives—sufficiently supported to justify undertaking that permissible residual risk?

The tests are sequential and jointly necessary: residual risk within limits does not establish that an activity remains worth undertaking, and high expected value cannot make an inadmissible activity acceptable.

Table 1. RWR three-test authorization sequence

Test	Plain-language question	If not satisfied
Admissibility	May this action proceed at all?	Do not authorize.
Risk permissibility	Is residual risk within permitted limits for the proposed conditions?	Do not authorize in the proposed form.
Value justification	Does the current value case justify undertaking that permissible risk?	Do not authorize; adapt, narrow, or reconsider.

5.2 Value justification as governed judgment

Value justification need not convert every outcome or value dimension into a universal scalar. Financial, operational, customer, workforce, distributional, governance, strategic, and learning outcomes can use different units, affect different stakeholders, and require different complementary inputs. The authorization body therefore makes a governed judgment from explicit evidence about outcomes, value, costs, and alternatives instead of relying on a mandatory aggregate score.

At minimum, the decision record should identify the current purpose, principal beneficiaries and burden-bearers, material outcome dimensions, the value those outcomes are expected or observed to create, material resource requirements and costs, feasible alternatives, unresolved uncertainty, and the scope of action being justified.

Evidence strength should also be explicit. Evidence may be insufficient, directional, decision-sufficient, contradictory, nonportable, or obsolete. The standard can vary with the consequence of the decision: evidence adequate for a restricted pilot need not be adequate for enterprise expansion.

5.3 Least necessary permissible exposure

The guiding principle is satisficing, not universal optimization. The authorization body first excludes inadmissible actions, then actions whose residual risk exceeds authorized limits, and then actions whose value case is insufficient for the decision. Among the remaining feasible alternatives that can pursue the justified purpose, it prefers options that avoid unnecessary residual risk. This approach does not assume that heterogeneous risks can always be reduced to a single ordered quantity.

5.4 Alternatives remain live

Feasible alternatives are not only an initial business-case input. They remain part of continuing justification, but they need not be fully re-underwritten at every review. Comparator analysis is a governed judgment using the same evidentiary discipline as value justification, not a hidden aggregate score. Alternatives should be refreshed when new evidence emerges, their costs, required inputs, expected value, or risk exposure change materially, or the value case itself shifts. An activity should not remain justified merely because it produces some favorable outcomes if a feasible approach can pursue sufficiently comparable value with materially lower cost, fewer required resources, less exposure, or stronger evidence.

For a specific AI activity, the practical implication is that authorization remains relative to the alternatives that are genuinely feasible at the current decision point, not only to the status quo considered at initial approval.

5.5 Material change and reauthorization

For an authorized AI activity, a change is material when updated conditions place the outcome of admissibility, risk permissibility, or value justification in genuine doubt—so that re-running the three tests could plausibly change the authorization decision. Potential triggers include changes in purpose, principal beneficiary, stakeholder distribution, comparator, outcome dimension, decision horizon, scale, operating conditions, evidentiary support, risk exposure, or a non-compensable constraint. Material change can trigger reauthorization at any point, including during a planned learning stage. A revised rationale is not necessarily improper; when material, it is a new basis for accepting residual risk and should be treated accordingly.

For example, if a document-preparation AI is re-scoped to recommend credit decisions, the activity has changed materially even if the underlying vendor or model has not.

6. Governing Concurrent Learning

6.1 The operating cycle

When important uncertainty remains but can be reduced safely, RWR supports a staged operating cycle. Figure 1 summarizes the sequence from specifying purpose and directing inquiry through controlled action, evidence updating, and reauthorization or adaptation.



Figure 1. RWR operating cycle

The sequence describes the default rhythm of controlled learning; a material-change trigger can interrupt it and require reauthorization before the planned stage ends. Acting is appropriate only when unresolved uncertainty can be reduced through limited action, the observation can inform a decision, and the exposure required to learn is itself admissible and risk-permissible. The learning stage must also satisfy value justification: the expected decision value of learning must be sufficient to justify undertaking the permissible exposure.

6.2 Learning object

A claimed learning benefit should be specified prospectively. At minimum, the organization should identify:

- the decision-relevant question or proposition to be tested;
- the measure and baseline or comparator;
- the evidence horizon and decision criterion; and
- what the result would change in the next authorization decision.

Information generation alone is not a sufficient benefit. Learning matters when it can change an identified action.

6.3 Separate evidence, coupled decisions

The organization governing the AI activity should maintain distinguishable evidence about value and risk even when the same data sources contribute to both. Strong control evidence does not prove value, and strong value evidence does not repair a weak risk case. Independence here refers to the validity of the judgments, not necessarily to physically separate data systems. Shared evidence can be used, but ownership, interpretation, challenge, and decision consequences should remain explicit.

The two evidence streams interact through governance. Evidence about realized outcomes and value may expose complementary inputs or dependencies requiring new risk inquiry. Evidence about emerging risk may change the feasibility, cost, distribution, or desirability of the value pathway. Neither stream determines the other; both affect authorization.

6.4 Institutional form

RWR is intended as a decision protocol that can be layered onto existing governance rather than as a requirement for a new institution. The authorization forum must possess or obtain the capabilities needed to evaluate both cases: explicit value-case ownership and challenge, independent risk assessment, relevant legal/compliance and technical input, authority to condition or withdraw authorization, and a documented escalation path when material disagreements cannot be resolved. The organizational form may vary, and the depth of documentation, challenge, and review should scale with the consequence and uncertainty of the activity.

7. Aster Vale Revisited: A Governance Decision

7.1 Original purpose and alternatives

Navigator's original value proposition combined expected operational, customer, and access value. AI-assisted preparation was expected to reduce underwriter work, identify missing information earlier, accelerate completed files and decisions, reduce abandonment, and make individualized review more economical for smaller applicants. The bank retained five alternatives: the existing process; non-AI workflow redesign; conventional automation; a narrower AI-assisted preparation tool; and the Navigator pilot.

That expected value pathway directed inquiry into whether review and correction costs offset saved time, whether nonstandard applicants experienced incorrect information requests, whether document

preparation was actually the binding delay, whether faster decisions affected abandonment, whether source access supported human verification, and whether pilot controls could survive scale.

7.2 What the bank learned

The table below summarizes what Aster Vale learned during the restricted Navigator pilot. The assessment labels are qualitative reading aids rather than points on a common scale: some describe evidence strength (for example, directional or insufficient), while others describe the pattern of observed effects (for example, mixed or uneven).

Dimension	Pilot observation	Assessment
Financial	Preparation cost falls, but vendor, validation, QA, and integration absorb much of the saving.	Directional
Operational	Preparation time falls 28%; application-to-decision time falls 7%.	Mixed
Customer	Repeated requests fall 4%; abandonment changes from 13.0% to 12.7%.	Insufficient
Access	No attributable approval effect; weak improvement for standardized small applications.	Insufficient
Distribution	Standard digital applications improve; nonstandard-document processing time rises 6%.	Uneven
Workforce	Experienced users save time; verification burden persists; junior reliance increases.	Mixed
Governance	Draft consistency improves; source accessibility falls from 94% to 64%.	Mixed / weakened
Learning	Downstream bottlenecks, user differences, document limits, and scaling conditions become clearer.	Supported for pilot
Comparative	Non-AI redesign reduces preparation time 17% and decision time 6%.	AI advantage limited

Navigator therefore produces credible preparation-efficiency and learning outcomes, but the evidence that those outcomes translate into customer, access, or broad financial value remains weak. Some effects are uneven, and the non-AI alternative delivers similar end-to-end improvement with less technological complexity. The decision logic depends on this pattern—clear task-level improvement, weaker downstream value, a credible alternative, changing risk evidence, and limited portability from pilot to scale—rather than on any single numerical threshold.

7.3 The revised request

Aster Vale management proposes enterprise expansion, larger loans, lower quality-assurance coverage, broader documents, and less-experienced users. The rationale shifts toward processing capacity, avoided hiring, and strategic AI capability. Both cases therefore change: the principal value rationale and

comparator shift while exposure, user heterogeneity, consequence, and uncertainty increase. Pilot evidence becomes less portable at the moment management seeks to rely on it more heavily.

7.4 What different governance perspectives ask

The Aster Vale expansion decision can be viewed through several governance perspectives, each emphasizing a different question. Table 2 compares those decision lenses and shows where RWR places its specific emphasis.

Table 2. Governance perspectives on the Aster Vale expansion decision

Perspective	Primary question in the Aster Vale decision
Risk management	Is residual risk acceptable for the proposed scope and controls?
Benefits realization	Are the expected benefits being realized and owned?
Adaptive management	What uncertainty should the next stage reduce, and how should action adapt?
PRAIG	What organizational posture should manage the persistent tension between AI value creation and risk mitigation?
RWR	Do the current value case and current risk case, relative to feasible alternatives, support the same proposed action?

These perspectives can complement one another. OCC guidance, for example, could support a similar ultimate disposition [1]. RWR focuses more narrowly on the relationship between the two evolving cases at the point of authorization. When the value rationale changes materially, the basis for accepting residual risk changes with it and should be reconsidered.

7.5 Disposition

Restricted preparation use remains admissible and conditionally risk-permissible. Full expansion is not yet established as risk-permissible because source-access deterioration, user reliance, document heterogeneity, vendor dependency, and reduced proposed oversight change the exposure. The original customer-and-access value case is only partially supported, while the revised efficiency-centered case has not been formally challenged against total costs and the non-AI alternative.

RWR therefore supports adaptation and reauthorization rather than automatic expansion, abandonment, or unchanged continuation. A second controlled stage would restore source access, retain scope limits, compare Navigator directly with workflow redesign, include total implementation and control costs in the value case, examine distribution and workforce effects, and specify what evidence would justify broader use.

The Aster Vale illustration shows how RWR can change the framing of a continuation or expansion decision. We now turn to what the approach adds, the institutional conditions it requires, and the questions that remain open.

8. Discussion

8.1 What RWR adds

RWR contributes an activity-level, evidence-triggered authorization discipline for situations in which value and risk are learned concurrently. It is process-determinate rather than outcome-determinate: the framework specifies what must be considered, by whom, in what sequence, with what evidentiary record, and what triggers reconsideration. Competent decision makers may still reach different judgments. The empirical question is whether this discipline changes what organizations notice, document, challenge, or decide.

The contribution should therefore be judged by a practical research question: does making this relationship explicit cause organizations to notice different dependencies, demand different evidence, challenge purpose changes earlier, or make different continuation and scaling decisions? The present paper proposes the architecture; it does not answer that empirical question.

8.2 Why now

The practical significance of RWR follows from the decision environment developed in Sections 1 and 2. AI activities can change in purpose, configuration, users, alternatives, and evidence while organizations are still learning what value and risk actually materialize. RWR is designed for that condition; it does not depend on a claim that pre-AI governance was mistaken.

8.3 Institutional implications

Implementation requires less institutional novelty than procedural discipline. At minimum, organizations need named ownership and versioned updates for the value and risk cases; a credible value-challenge capability; an authorization forum empowered to condition, restrict, expand, suspend, or terminate the activity; decision-specific evidence expectations; proportionate review depth; and escalation when material disagreement persists. These are minimum capabilities rather than prescribed committee seats or universal templates.

Monitoring should connect indicators to the value pathway and the risk case rather than placing generic benefit and risk dashboards side by side. The organization should know which intermediate outcomes must occur, how those outcomes are expected to create value for specified stakeholders, what complementary inputs and costs matter, what evidence supports attribution, which assumptions remain unresolved, and what observations would change the next authorization decision.

8.4 Scope beyond AI

The underlying principle may also apply in domains characterized by incomplete and evolving understanding of both purpose and consequence, including emerging technologies, healthcare, infrastructure, environmental adaptation, and public experimentation. We do not claim general validity across those settings. AI is the focal application because its components, uses, deployment conditions, and organizational meaning can change rapidly and simultaneously.

8.5 Limitations and research agenda

RWR has several limitations in its present form. First, the framework is conceptual and has not yet been tested in live organizational decisions, so we do not know whether it materially improves authorization outcomes or simply adds governance effort. Second, value justification remains a governed judgment; RWR structures the evidence and decision process but cannot eliminate disagreement over how competing forms of value should be weighed. Third, the framework assumes that organizations can maintain a sufficiently independent risk assessment while allowing the value case to shape additional inquiry, an institutional separation that may be difficult in practice. Fourth, keeping alternatives visible and reopening authorization after material change can impose meaningful information and coordination costs. Finally, the Aster Vale illustration shows how the logic can operate in one constructed setting but cannot establish how well the approach travels across sectors, organizational sizes, or higher-consequence AI uses.

Further work should test and refine RWR. Relevant questions include:

Risk discovery. Does value-directed inquiry identify different or earlier risks than category-based assessment alone?

Decision effects. Does explicit separation of risk permissibility and value justification change continuation or scaling decisions?

Persistence and burden. Does formal reauthorization after purpose or value revision prevent weak projects from persisting, or merely add procedural burden?

Institutional design. What institutional arrangements produce credible value challenge without transferring business judgment to risk functions?

Learning versus delay. Under what conditions does concurrent learning justify staged action rather than delay or termination?

A near-term validation step could use a structured expert or practitioner review. Participants could assess the same AI-use case under their ordinary governance practice and then under the RWR three-test protocol, with differences in requested evidence, identified dependencies, rationale, and disposition recorded. This would not establish effectiveness, but it would test whether RWR changes decision-relevant reasoning before more costly comparative field studies are attempted.

9. Conclusion

Organizations increasingly deploy AI before they fully understand which outcomes an activity will produce, how much value those outcomes will create, what resources and costs will be required to realize that value, or what risks the activity will generate. The governance challenge is therefore not only to assess risk and monitor benefits, but to decide whether evolving evidence about the value case and risk case continues to support the same organizational action.

We propose RWR as one way to organize this continuing decision. Independent risk identification provides the coverage floor, intended value adds purpose-directed emphasis, and the three authorization tests keep permissibility and value justification distinct. Alternatives remain visible, and material change can reopen the decision. The practical question is simple: given what we know now, should this activity continue in its current form?

Whether RWR improves governance remains an empirical question. This paper defines the approach and the problem it is intended to address; the next step is to test whether it changes decision-relevant reasoning in practice.

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